Log Analysis: Automated Error Clustering & Signature Libraries

Duration : 90 Min.

Scenario

You need to write a Python script that reads raw application logs, normalizes stack trace signatures, clusters identical errors, and matches them to runbook entries.

Target File Location & Creation

File to Create/Update: `student_workspace/cluster_errors.py` **Input Resource File to Inspect**: `student_workspace/system_errors.log`

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Inspect `system_errors.log` in `student_workspace/`. - Edit `cluster_errors.py` and record the required log analytics or findings.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect `system_errors.log` inside `student_workspace/`.
3. Open `cluster_errors.py` in `student_workspace/` and perform the required modifications.
4. Save your changes (`Ctrl + S` or `Cmd + S`).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running `python run.py` locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	cluster_errors.py is syntax-valid Python	4 Marks

TC2	Parses error logs and normalizes dynamic IDs/timestamps	4 Marks
TC3	Groups identical error signatures into clusters	4 Marks
TC4	Writes structured JSON output to error_clusters.json	4 Marks
TC5	Matches cluster signature to runbook ID	3 Marks
TC6	Calculates error frequency per cluster	2 Marks
TC7	Handles missing/malformed log lines gracefully	2 Marks
TC8	Includes first and last occurrence timestamps	2 Marks

Total Score: 25 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.